

ZHIYU SUN

zs2710@columbia.edu | 646-552-5444 | zhiyusun.com

SKILLS

- Languages: Python, C++, MATLAB, C, TypeScript, Bash, SQL, C#, R, LaTeX
- Libraries: NumPy, TensorFlow, Pandas, Matplotlib, Qt, PsychoPy
- Tools: Git, SolidWorks, Simulink, VS Code, Bambu Studio, Power BI

PROFESSIONAL EXPERIENCE

VisionNav Robotics, Inc.

Atlanta, GA

Senior Project Engineer

Apr 2026 – Sep 2026

- Delivered 3 AGV automation solutions for leading global manufacturers, incorporating PLC integration, task scheduling, and route planning to achieve throughput exceeding 14 loads/hour.
- Handled on-site commissioning independently, coordinating remotely with 3 cross-functional teams to resolve technical issues and translating customer requirements into technical solutions.
- Calibrated AGV fork positioning, steering, and LiDAR sensors; validated point cloud alignment and configured laser safety zones for precise cargo handling, autonomous navigation, and obstacle avoidance.

SurgiVance, Inc.

New York, NY

Data Science R&D Intern

Oct 2025 – Feb 2026

- Built a TypeScript image annotation tool using OpenSeadragon and SVG overlays, applying coordinate transformations and event-driven synchronization to maintain consistent annotations across image views.
- Reduced label recalculation time by 60% through parallel heatmap loading and saving with ThreadPoolExecutor and GPU-accelerated segmentation mask computation with CPU/Numba fallbacks.

Guo Lab

New York, NY

Research Assistant

Feb 2025 – Jun 2025

- Implemented CPU-side data preloading and optimized Dice score computation to reduce GPU memory overhead and training latency in a CNN-Mamba model for 3D MRI segmentation.
- Reduced training time by 75% while maintaining a Dice score of 0.988.

Shanghai Microport Medbot (Group) Co., Ltd.

Shanghai, CN

Operations & Data Intern

Jun 2023 – Sep 2023

- Demonstrated robotic technologies to 50+ visitors daily across 4 exhibitions, securing 13 hospital-director contacts and translating customer feedback and clinical operation data into 2 team-adopted improvement plans.
- Developed a hierarchical, parameter-based data-sorting model to streamline inventory tracking and reduced inventory processing time by 50%.

PROJECTS

AI-Enabled Quadruped Robot with Vision and Audio Control

New York, NY

- Built a quadruped robot with custom 3D-printed leg assemblies, integrating a Raspberry Pi 4 with a servo controller and daisy-chained bus servos for joint actuation.
- Integrated Gemini and YOLOv5n for visual perception, Porcupine for wake-word detection, and Vosk for speech recognition to enable real-time voice control.

Cognitive Task Design in Python

New York, NY

- Designed a memory task presenting image stimuli and recording behavioral responses using PsychoPy.
- Integrated the task and real-time EEG recording into LSL for synchronized data acquisition.

Automated Financial Reporting System

New York, NY

- Built a financial reporting application integrating Power BI reports through REST APIs and automating Excel data ingestion, filtering, and table generation using Microsoft Access.
- Integrated DeepSeek-V3 to analyze filtered financial data and automatically generate PowerPoint reports.

EDUCATION

Columbia University

New York, NY

Master of Science in Biomedical Engineering

Sep 2024 – Dec 2025